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Soheil Kavian ; Arian Zarriz ; Matthew J. Powell-Palm 



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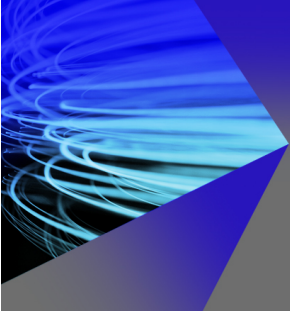
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Soheil Kavian,^{1,a)} Arian Zarriz,¹ and Matthew J. Powell-Palm^{1,2,3,a)}

AFFILIATIONS

¹J. Mike Walker'66 Department of Mechanical Engineering, Texas A&M University, College Station, Texas 77843, USA

²Department of Materials Science & Engineering, Texas A&M University, College Station, Texas 77843, USA

³Department of Biomedical Engineering, Texas A&M University, College Station, Texas 77843, USA

^{a)}Authors to whom correspondence should be addressed: soheilkavian@tamu.edu and powellpalm@tamu.edu

ABSTRACT

Modern industrial liquids—including coolants, lubricants, solvent blends, cryoprotectant cocktails, etc.—frequently employ complex, many-component formulations, but contemporary viscosity models and datasets are overwhelmingly limited to simplified binary or ternary compositions, leaving the most application-relevant compositional spaces broadly unexplored. This gap is attributable both to the limitations of classical viscosity correlations, which typically require either untenably idealized interaction assumptions or an untenably larger number of interaction parameters, and to the lack of systematic many-component viscosity datasets. Here, we provide a first-of-its-kind dataset of 44 316 viscosity measurements spanning 100 aqueous and organic solutions of up to 17-component complexity across temperatures from -20 to 35°C , and we use it to power a physics-informed neural network (PINN) model that provides unprecedented predictive power and physical insight into many-component solution viscosity. We first show that predictive implementations of two prominent classical models (Katti-Chaudhuri and augmented Adam-Gibbs) systematically fail to describe these solutions but retain valuable trend-level information. We then use these models as physical guides for our PINN, embedding classical model insight as a training aid while machine learning the residual non-ideal contributions that dominate in many-component composition space. The resulting model, which we also provide as a software application, maintains stable performance across solution complexity, outperforming both classical correlations and data-only artificial neural networks on fully withheld test data. Finally, we compare residuals of both the classical and PINN models as a function of component number and entropy of mixing, suggesting that entropic phenomena unaccounted for in classical models may dominate the viscosity of many-component solutions.

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NOMENCLATURE

Symbol description units

A_i	Arrhenius intercept for component i (–)
B_i	Arrhenius slope for component i (K)
f	fragility exponent (–)
L	total loss function (–)
$L1$	data fidelity loss (–)

$L2$	AAG physics residual (–)
$L3$	KC physics residual (–)
R	universal gas constant (J/mol K)
S_i	entropy coefficient of component i (–)
S_{mix}	solution configurational entropy factor (–)
T	temperature K or ($^{\circ}\text{C}$)
$T_{g,i}$	glass transition temperature of component i (K)
$T_{g,mix}$	solution glass transition temperature (K)
V_i	molar volume of component i (ml/mol)

V_{mix}	solution molar volume (ml/mol)
$W1, W2, W3$	loss weights (–)
W_{ij}	binary interaction parameter (J/mol)
w_i	mass fraction of component i (–)
x_i	mole fraction of component i (–)
η	viscosity (cP)
η_{∞}	high-temperature limiting viscosity (cP)
η_{AAG}	viscosity predicted by augmented Adam–Gibbs model (cP)
η_{KC}	viscosity predicted by Katti–Chaudhri model (cP)
η_g	viscosity at glass transition (Pa s)
η_i	viscosity of pure component i (cP)
Θ	reduced temperature T_g/T (–)
ϕ_i	volume fraction of component i (–)

INTRODUCTION

Viscosity is a fundamental transport property of fluids that governs flow behavior, mixing, and the transfer of momentum and energy. The viscosity of aqueous and organic liquids plays a dominant role in industrial and scientific domains ranging from chemical processing to advanced energy systems to organ and tissue cryopreservation.^{1–3} In practice, this viscosity dictates myriad processing parameters, whether pumping requirements, vascular or flow resistance, mixing efficiency, or the rates of both mass and heat transfer. As such, accurate prediction of the viscosity of organic solutions (both aqueous and otherwise) is indispensable in the design and use of those that serve as solvents,^{4,5} lubricants,^{6,7} cryoprotectants,^{8–11} and heat-transfer media.^{12,13}

While the challenge of solution viscosity prediction (both as a function of composition and temperature) has been met to excellent effect in binary and ternary aqueous and organic solutions of interest over the past several decades,^{4,14–17} the material realities of twenty-first-century industry increasingly operate over a much wider chemical and thermal parameter space, integrating many (>3) diverse components in solution in efforts to reach ever-greater performance across applications.^{18,19} The challenge of viscosity prediction (and property prediction writ large) intensifies significantly in many-component solutions, where the free-energy landscape becomes multidimensional,^{20,21} and diverse molecular interactions and non-idealities of mixing may lead to complex behavior that both defies analytical modeling and resists semi-empirical prediction given its massive compositional parameter space.

Classical viscosity prediction methods for liquid solutions rely on empirical and semi-empirical equations that link composition and temperature to viscosity through mixing rules.^{14,22} For composition dependence, commonly used mixing rules include the Arrhenius logarithmic mixing, Grunberg–Nissan, McAllister, and Katti–Chaudhri models,^{23,24} whereas temperature dependence is often captured by entropy-based theories such as the Adam–Gibbs framework.²⁵ While these frameworks are useful, they generally require system-specific fitting parameters that necessitate costly experimental data to prescribe and show poor generalization to multicomponent mixtures in which non-ideal interactions dominate.^{23,26}

In recent years, researchers have turned to machine learning (ML) to begin surmounting these challenges.¹² Methods such as artificial neural networks (ANNs), support vector machines (SVMs),

the group method of data handling (GMDH), and Gaussian process regression (GPR) have been applied to both pure liquids and relatively simple mixtures (2–3 components).²⁷ These approaches offer improvements over classical models by capturing nonlinear dependencies among viscosity, composition, and temperature without relying strictly on empirical mixing rules.¹² ANN- and SVM-based models, for example, have shown better accuracy than McAllister- or Arrhenius-type correlations in binary and ternary systems,^{27,28} while GMDH and GPR have extended applicability to alcohol–water, glycol, and glass-forming systems.²⁹ Recent works increasingly incorporate varying sources of intersecting chemical and physical information to enhance machine learning outcomes, ranging from physical descriptors generated via molecular dynamics³⁰ to molecular embeddings^{31–33} that enable simultaneous multi-property prediction. Although these studies demonstrate rapid progress in machine-learning-based viscosity modeling, their performance still depends heavily on the scope of available training data, and their generalization to many-component mixtures remains extremely limited, underscoring the need for versatile predictive frameworks able to grapple with significantly greater chemical complexity.²⁶

To meet this need, we here present twin advancements that extend state-of-the-art viscosity prediction to multicomponent solutions. First, we present a large experimental dataset of organic and aqueous solution viscosities, including 44 316 viscosity measurements across >100 solutions, ranging in complexity from 2 to 17 components, and sampled over a temperature span of –20 to 35 °C. Second, we present a physics-informed neural network (PINN) framework that incorporates augmented Adam–Gibbs theory and the Katti–Chaudhri model as physical guides to the training process, resulting in accurate temperature-dependent prediction of the viscosities of arbitrarily complex solutions of the 17 components investigated. Whereas prior viscosity models have been confined largely to binary (or at most ternary) systems, our framework achieves predictive capability for mixtures containing up to 17 components, marking a decisive step toward realistic modeling of complex organic solutions. Our results not only deliver substantially higher accuracy than both traditional correlations and purely data-driven ANN models but also expose the systemic inability of classical models to capture the physics of many-component solutions, offering intriguing clues into potentially fundamentally unaccounted-for physical phenomena affecting viscosity in these complex systems. Finally, we provide the complete raw dataset in a machine-readable tabular format in the [supplementary material](#) and a user-friendly implementation of our PINN model in an executable application for viscosity prediction.

CREATION OF MANY-COMPONENT TEMPERATURE-DEPENDENT VISCOSITY DATASET

Component selection and solution design

The first barrier to the creation of viscosity prediction tools for many-component (>3 component) solutions is the near-complete absence of many-component viscosity data in the scientific literature. As such, we chose 17 chemical components (water plus 16 commonly available organic molecules) and designed a collection of 100 solutions therefrom, comprising solutions with 2, 3, 5, 8, 9,

TABLE I. Pure components used in solution preparation and select molar properties.

Solutes	Atomic Mass (g/mol)	Density (g/cm ³)	Molar volume (cm ³ /mol)
Water	18.015	1	18.015
Methanol	32.04	0.792	40.45
Ethylene glycol	62.07	1.11	55.91
1,2-propanediol	76.09	1.04	73.16
1,3-butanediol	90.12	1.0053	89.64
Diethylene glycol	106.12	1.12	94.75
N,N-dimethylethanolamine	89.14	0.887	100.49
2-methyl-2,4-pentanediol	118.17	0.923	128.02
Triethylene glycol (TER)	150.17	1.12	134.08
Diethylene glycol monobutyl ether	162.23	0.954	170.05
Glycerol	92.09	1.261	73
DMSO	78.13	1.1	71
3-methoxy-1,2-propanediol	106.12	1.114	95.3
1,2,4 butanetriol	106.12	1.19	89.2
1,3-propanediol	76.09	1.059	71.8
2,3-butanediol	90.12	0.99	91.3
1,4-butanediol	90.121	1.017	88.61

10, 12, 15, and 17 components; the viscosity of each was measured between 35 and -20°C (phase transition permitting).

Molecules were chosen based on four criteria: (1) melting point $<20^{\circ}\text{C}$ (i.e., liquid at room temperature); (2) broad commercial availability (cost $<\$0.4/\text{g}$); (3) plausible relevance to heat exchange, chemical processing, or other viscosity-limited applications; and (4) relevance to cryopreservation (i.e., non-acutely toxic and membrane permeable, as evaluated by Ahmadkhani *et al.*¹⁰). These 17 liquids span a wide range of hydrogen-bonding capabilities, polarity, and molecular size (i.e., molar volume), offering a chemically diverse basis for probing viscosity across composition and temperature. The complete list of pure components used, alongside select molar properties, is presented in Table I, with values reported at 25°C . These data were obtained from Sigma-Aldrich and Thermo Fisher technical data sheets and were verified against NIST databases.³⁴

Pure-component viscosities were first measured for each, in good agreement with the literature (where available). Solutions were generated by randomly selecting components and mole fractions for each of the desired component numbers (under the constraint that the sum of mole fractions equaled unity), ensuring broad and unbiased coverage of compositional space. The full list of solutions tested, along with their measured temperature-dependent viscosities, is provided in [supplementary material 2](#).

Viscosity measurement

Viscosity measurements were performed using a ViscoLab 3000 viscometer (Cambridge Viscosity) coupled to a calibration-grade Fluke heating/cooling circulating bath (shown schematically in [supplementary material](#) Fig. S1). Approximately 2 ml of sample was loaded into the test chamber, alongside a calibrated piston selected for the anticipated viscosity range, the oscillation of which provides the working principle for electromagnetic viscometry. The bath was programmed via MATLAB through RS-232 serial

communication to step the temperature from -20 to 35°C . At each setpoint, viscosity was recorded after thermal equilibration using the viscometer's internal averaging algorithm. The rated system accuracy is $\pm 1\%$ of full scale for most piston ranges ($\pm 5\%$ for the highest range), with a rated repeatability of $\pm 0.8\%$ of reading. We conducted benchmark calibration tests using Koehler S600, a hydrocarbon-based fluid standard widely adopted for viscometer validation. The S600 standard provides NIST-traceable viscosity-temperature values (according to ASTM D2162) and features a smooth, monotonic response to temperature.³⁵ Calibration outcomes ([supplementary material](#) Note 10) demonstrated excellent agreement between measured and certified values, confirming the absence of systematic biases. In addition, a comprehensive uncertainty analysis based on JCGM 100:2008 (GUM) was performed³⁶ for all data acquired; the combined expanded uncertainty is summarized in [supplementary material](#) Note 8 and typically ranges from 1% to 3%, depending on temperature and piston range. These calibration and uncertainty assessments validate the reliability of the dataset used for model development and validation. A detailed description of the platform, experimental protocol, calibration, and error analysis is provided in [supplementary material](#) 1. Histograms showing distributions of viscosities and temperatures represented in the dataset are provided in [supplementary material](#) Note 12 therein.

DEVELOPMENT OF AN OPTIMAL MODEL FOR VISCOSITY PREDICTION IN MANY-COMPONENT SOLUTIONS

Armed with an unprecedented viscosity dataset for many-component solutions, we proceeded to iteratively apply or develop multiple predictive models that could be validated against or trained with this dataset to enable prediction of temperature-dependent viscosity for arbitrarily complicated solutions of the underlying 17 components.

We began by evaluating two popular physical models used to predict compositional and temperature dependences of viscosity: the idealized Katti–Chaudhuri (KC) model and an augmented Adam–Gibbs model (AAG). In light of their limited accuracy, we proceeded to test two Artificial Neural Network (ANN) models. Finally, to achieve optimal accuracy and extensibility, we combined the analytical physical insight encoded by the classical models with the empirical power of the ANN models to develop a physics-informed neural network (PINN model), which outperformed both ANN and classical mixing rules across solution compositions and temperatures. We will first describe each model tested, then compare their relative predictive powers.

Physical model 1: Katti–Chaudhuri (KC) model

The KC model describes viscosity as an activated process, extending Eyring's theory of viscous flow to solutions by assuming a regular-solution form for the activation free energy. For each pure component i , viscosity is prescribed an Arrhenius-like expression with temperature:^{37,38}

$$\ln \eta_i(T) = A_i + \frac{B_i}{T}, \quad (1)$$

where A_i and B_i are obtained by regression of experimental viscosity data against reciprocal temperature. For an n -component solution, the KC model then relates the viscosity of the solution η_{KC} to the properties of its constituents according to

$$\ln(\eta_{KC} V_{\text{mix}}) = \sum_{i=1}^n x_i \ln(\eta_i V_i) + \frac{1}{RT} \sum_{i < j} x_i x_j W_{ij}, \quad (2)$$

where x_i is the mole fraction, V_i is the molar volume of component i , $V_{\text{mix}} = \sum x_i V_i$, R is the universal gas constant, and W_{ij} is a binary interaction energy for viscous flow. The incorporation of the interaction parameters W_{ij} , while useful for the descriptive application of the model to binary solutions, significantly hinders its predictive use, as it requires experimental data for the specific combination of components of interest. For many-component solutions, characterization of the many possible binary (and higher-order) interaction parameters becomes rapidly intractable. As such, we adopted the ideal form of the model (in which the interaction term is set to zero), which still enables true prediction of solution viscosity (i.e., prediction based only on pure component properties):

$$\ln(\eta_{KC} V_{\text{mix}}) = \sum_{i=1}^n x_i \ln(\eta_i V_i). \quad (3)$$

The assumption of ideality reduces the model to a composition-weighted logarithmic average of pure-component contributions. We note, furthermore, the analogy of this form to both the ideal and size-dependent (e.g., Flory-style) entropies of mixing used in classical theories of solution thermodynamics, which may be interpreted, to a limited degree, to suggest the link between solution entropy and viscosity posited by Adam, Gibbs, Bell, and others.^{25,39} Details of the regression procedure used to obtain A_i , B_i , and the full implementation are provided in [supplementary material](#) Note 2, with fitted parameters for all 17 components listed in Table SI 1.

Physical model 2: Augmented Adam–Gibbs (AAG)

The Adam–Gibbs framework connects the rapid, super-Arrhenius increase in viscosity near the glass transition to a decrease in configurational entropy. We use an augmented form that replicates Arrhenius behavior at high temperatures while capturing the sharp increase as T approaches T_g , and we evaluate viscosity with the following equation:^{26,40}

$$\eta_{\text{AAG}}(T, w_i) = \eta_{\infty} \exp \left[\ln \left(\frac{\eta_g}{\eta_{\infty}} \right) \frac{\Theta}{s_{\text{mix}} - (s_{\text{mix}} - 1)\Theta^f} \right], \quad (4)$$

where w_i is the mass fraction of compositions, η_{∞} is the viscosity in the limit of high temperature, η_g is the viscosity at T_g , and f is a fragility exponent. For an n -component mixture, s_{mix} , $T_{g,\text{mix}}$ and Θ are defined as follows:²⁶

$$\Theta = \frac{T_{g,\text{mix}}}{T}, \quad (5)$$

$$s_{\text{mix}} = \sum_{j=1}^n w_j s_j, \quad (6)$$

$$T_{g,\text{mix}} = \sum_{j=1}^n w_j T_{g,j}. \quad (7)$$

Here, w_j is the mass fraction of each pure liquid, s_j is the coefficient related to the configurational entropy of each pure liquid, and $T_{g,j}$ is their glass-transition temperature. Glass transition temperatures of the pure components were either measured via differential scanning calorimetry (DSC) or drawn from literature (values provided in Table S2).

In contrast to the KC model, which can be applied in a fully predictive, idealized form using only pure-component parameters, the AAG framework requires additional solution-specific inputs (η_g , η_{∞} , f , and the component-specific entropy coefficients s_j), which require additional calibration. To enable predictive application across all 17-component mixtures, we held η_{∞} and f constant (consistent with prior observations for chemically similar systems²⁶), and estimated both these parameters and the s_j values using particle swarm optimization applied only to a restricted subset of the experimental dataset (the same training subset used for the machine-learning models). We fixed η_g at 10^{12} Pa s, consistent with common definitions of the glass transition.⁴⁰ Full details of the fitting procedure and implementation are provided in [supplementary material](#) Note 3. These idealizations inevitably reduce accuracy but allow the AAG to serve as a broadly extensible, semi-empirical predictive model for the purposes of predictive model comparison.

Artificial neural network (ANN) models

Given the unsatisfactory capacity of the KC and AAG models to enable the desired accuracy of viscosity prediction, we proceeded next to two artificial neural network (ANN) models. In the first model (ANN 1), each sample was represented by a feature vector of 18 dimensions (17 mole fractions plus temperature), and in the second (ANN2), each was represented by a feature vector of 35 dimensions (17 mole fractions and corresponding mass fractions plus temperature, encoding component molar mass), with the target

variable in each case being the natural logarithm of viscosity in units of centipoise (cP). We note that we trialed target variables of absolute (linear scale) viscosity and base-10 logarithmic viscosity, and we found the natural logarithm of viscosity to be most effective. To eliminate scaling disparities, all features and outputs were normalized to $[0,1]$. Data were divided into 70% training, 10% validation, and 20% testing subsets, with the test set limited to fully withheld solutions not seen during training or validation.

To develop the training dataset, we initially mapped all possible solute combinations (in solutions of 1–17 components), then selected 80 representative combinations with the goal of broadly sampling solution complexity and chemical variety. Equimolar solutions were then formulated from these combinations, and in cases where select solutes appeared to remain underrepresented, we incorporated slightly perturbed compositions derived from the same component combinations. The final training dataset thus comprises equimolar solutions supplemented with randomized variants to enhance robustness and mitigate extrapolation errors. We note that our baseline choice of equimolarity was made to ensure we sample the largest possible entropy-of-mixing space and capture difficult-to-predict, interaction-rich solutions. The independent test dataset comprised 20 fully withheld mixtures, with five test solutions generated randomly for each of four solution complexities (2-component, 5-component, 10-component, and 17-component solutions).

Crucially, none of these test compositions or their nearby variants appeared in the training or validation data, providing a rigorous test of the model's extrapolative capabilities.

Neural networks can be trained using various algorithms, including stochastic gradient descent, resilient backpropagation, conjugate gradient methods, and quasi-Newton approaches.⁴¹ Here, training was performed with the Adam optimizer, which provided stable convergence across architectures.⁴² Different hidden-layer sizes and activation functions were tested, as shown in Tables S4 and S5. The final setup for both ANNs was a feedforward neural network with one hidden layer (an MLP architecture), consisting of 40 neurons for ANN1 and 30 for ANN2, each with a ReLU activation function.⁴² Complete implementation details and hyperparameter tuning are included in [supplementary material Note 4](#) (ANN1) and [supplementary material Note 5](#) (ANN2). The incorporation of mass fractions in ANN2 notably enhanced correlation relative to ANN1 within the training dataset, though it provided only a marginal benefit in the downstream testing data.

Physics-informed neural network (PINN)

Finally, in an effort to improve accurate model extensibility outside of the training data and realize the goal of optimal many-component viscosity prediction, we married the physical

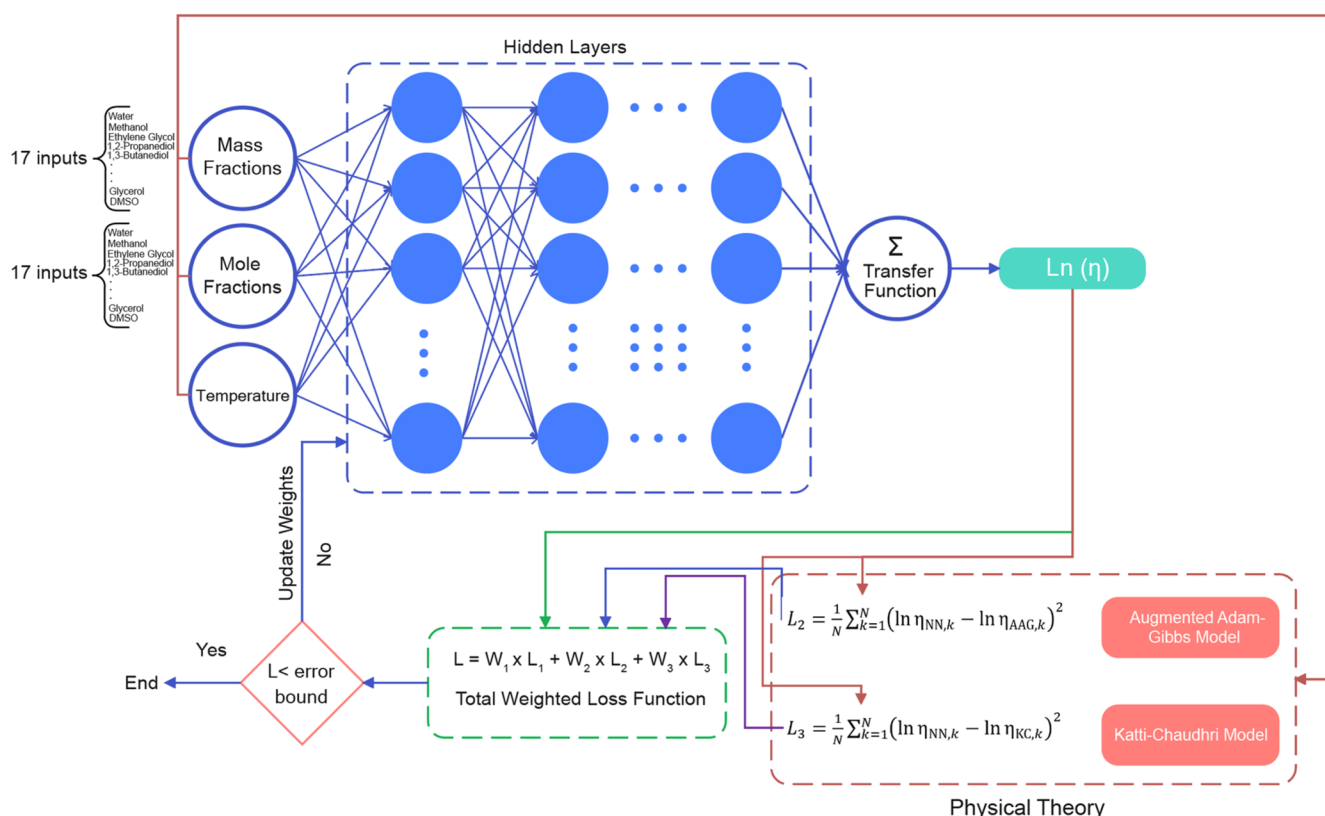


FIG. 1. Architecture of the PINN integrating data and physical models for viscosity prediction. The PINN combines experimental data with physical constraints from the augmented Adam–Gibbs and Katti–Chaudhri models to predict $\ln(\eta)$ from mixture composition and temperature. The total loss integrates data-driven and physics-based terms, ensuring predictions remain consistent with thermodynamic theory.

models and machine learning models above within a single physics-informed neural network (PINN). PINNs incorporate analytical physical theory into neural network loss functions to provide physically reasonable guidance for model training and prediction, and they have been shown to significantly enhance training efficiency and model extrapolability across a wide range of physical prediction scenarios.^{43,44}

Here, we incorporate the two physical models tested above (the ideal KC model and the AAG model) as physics-based constraints, in addition to the standard data-driven loss. The total loss was defined as a weighted combination of data loss and physics-informed residuals:

$$L = W_1 L_1 + W_2 L_2 + W_3 L_3, \quad (8)$$

where L_1 represents the mean-squared error between experimental and predicted log-viscosity, L_2 is the residual from the augmented Adam–Gibbs model, and L_3 is the residual from the ideal Katti–Chaudhuri model, and W_1 , W_2 , and W_3 are tunable weights. The data fidelity term was defined as the mean-squared error between experimental and predicted log-viscosity:

$$L_1 = \frac{1}{N} \sum_{k=1}^N \left(\ln \eta_{\text{exp},k} - \ln \eta_{\text{NN},k} \right)^2. \quad (9)$$

The AAG loss compared the network prediction with the mixture viscosity given by the augmented Adam–Gibbs relation, where $\eta_{\text{AAG},k}$ is given by Eq. (4):

$$L_2 = \frac{1}{N} \sum_{k=1}^N \left(\ln \eta_{\text{NN},k} - \ln \eta_{\text{AAG},k} \right)^2. \quad (11)$$

The KC loss enforced consistency with the Katti–Chaudhuri logarithmic mixing rule:

$$L_3 = \frac{1}{N} \sum_{k=1}^N \left(\ln \eta_{\text{NN},k} - \ln \eta_{\text{KC},k} \right)^2, \quad (12)$$

where $\ln \eta_{\text{KC}}$ is drawn from Eq. (3), such that

$$\ln \eta_{\text{KC}}(T, x) = \sum_{i=1}^n x_i \ln \left(\eta_i(T) V_i \right) - \ln \left(\sum_{i=1}^n x_i V_i \right). \quad (13)$$

The network simultaneously minimized prediction error against experimental data and ensured consistency with the governing physical models, as schematically depicted in Fig. 1.

The design of the PINN architecture was guided by the recommendations of Raissi *et al.*,⁴⁵ who suggested using deep architectures with multiple blocks and moderate neuron counts for physics-regularized training. In our case, the final architecture consisted of seven fully connected blocks, each with 20 neurons. Several activation functions were tested, and the sigmoid function produced the best performance.

Different weightings of the physics-informed loss terms and variations in the number of blocks were manually explored, with results summarized in [supplementary material](#) Note 6 (Table S6). Optimal performance was achieved by selecting weights that balanced data fitting and physical consistency, enabling the PINN to match the ANN performance on the training set while demonstrating lower error on unseen test mixtures.

COMPARATIVE MODEL PERFORMANCE

The performance of the physics-informed neural network (PINN) was benchmarked against two purely data-driven artificial neural networks (ANN: Method 1 and Method 2), as well as the augmented Adam–Gibbs (AAG) and Katti–Chaudhuri (KC) models. All models were trained using $\ln(\text{viscosity})$, while performance metrics and figures are reported using $\log_{10}(\text{viscosity})$ for readability and direct comparison to conventional viscosity data ranges. Figure 2 shows parity plots of predicted vs true logarithmic viscosities, and Table II reports the statistical metrics. All error metrics (RMSE, MAE) are reported in $\log(\text{cP})$ units.

On the unseen test mixtures, the PINN achieved an RMSE of 0.0709 $\log(\text{cP})$, compared to 0.0754 for ANN Method 2 and 0.0765 for ANN Method 1, representing a ~6% to 7% improvement in predictive error over the ANNs. In contrast, the AAG model showed a much higher error of 0.273, meaning the PINN reduced test-set RMSE by about 74% relative to AAG. The KC model performed even worse, with an RMSE of 0.427, indicating that the PINN reduced error by more than 83% compared to KC. Critically, the PINN shows a slightly higher training-set RMSE than ANN 2 (0.0386 vs 0.0248), even while providing lower test-set error. This represents the hallmark behavior of physics-informed models—unlike purely data-driven ANNs, which may overfit the training data, the PINN incorporates physical insight to penalize predictions that violate known scaling or mixing trends. These constraints intentionally limit the hypothesis space, preventing overfitting even at the cost of a small increase in training error. As a result, the PINN generalizes better to unseen mixtures, despite a looser training fit.

The relatively weak performance of both standalone AAG and KC models arises from their idealized assumptions, which overlook explicit intermolecular interactions such as hydrogen bonding and cooperative multi-body effects that greatly influence viscosity in multicomponent mixtures. When used alone, these models cannot capture such effects and therefore have limited predictive power. However, within the PINN framework, they act as physics-based regularizers: the network prediction is constrained to adhere to the overall thermodynamic structure of the AAG and Katti relations, while the ANN component adjusts its weights to implicitly account for the missing multicomponent interactions. This hybrid approach allows the PINN to preserve correct global physical trends while enhancing accuracy in extrapolated regions beyond the training data. The results show that, on the test set, the PINN not only achieved a lower RMSE compared to ANNs but also significantly outperformed the classical AAG model, which deviated considerably from experimental values.

Retention of PINN accuracy across solution complexity and implications for physical models

To further investigate the predictive performance of the developed PINN model, and to examine the limitations of the physical insight offered by the idealized physical models employed, we proceeded to evaluate the accuracy of each model as a function of solution complexity, which we quantified both in terms of the number of components and the calculated entropy of mixing.

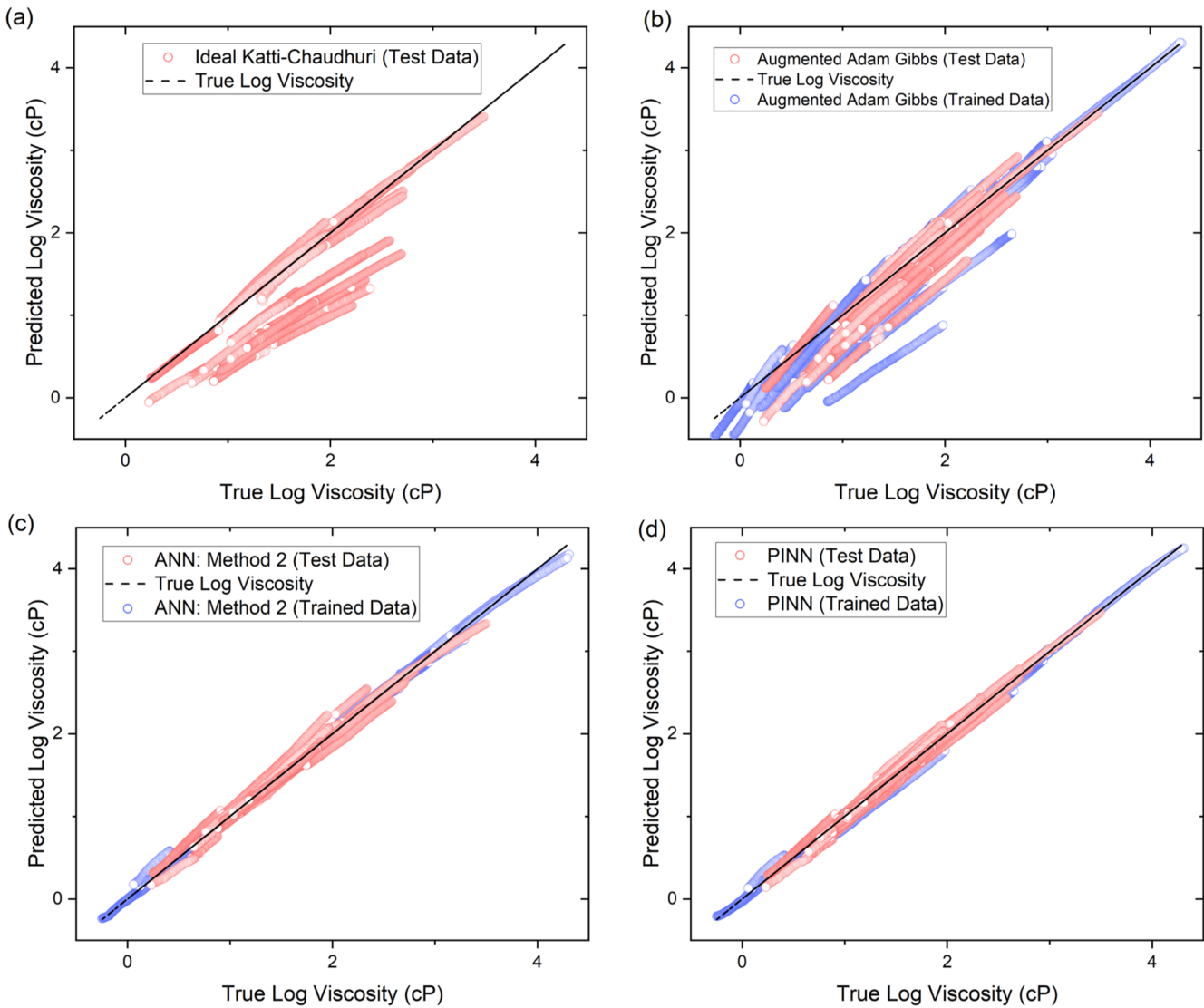


FIG. 2. Comparison of predicted vs experimental viscosities across models, displayed as log(cP). Models include (a) the ideal Katti–Chaudhuri model, (b) the augmented Adam–Gibbs (AAG) model, (c) ANN Method 2, and (d) the physics-informed neural network (PINN), evaluated across training and test datasets.

TABLE II. Metrics for training and test datasets for different ML models.

	Training set			Test set		
	R^2	RMSE [$\log_{10}(\text{cP})$]	MAE [$\log_{10}(\text{cP})$]	R^2	RMSE [$\log_{10}(\text{cP})$]	MAE [$\log_{10}(\text{cP})$]
PINN	0.9972	0.0385	0.0270	0.9852	0.0709	0.0508
ANN: Method 2	0.9988	0.0248	0.0161	0.9833	0.0754	0.0582
ANN: Method 1	0.9975	0.0361	0.0240	0.9828	0.0764	0.0591
AAG	0.8830	0.2496	0.1851	0.7812	0.2732	0.2259
KC	...	...	...	0.6948	0.4266	0.2999

In Fig. 3, log residuals [$\log(\eta_{\text{pred}}) - \log(\eta_{\text{exp}})$] and absolute percent error ($APE = \left| \frac{\eta_{\text{pred}} - \eta_{\text{exp}}}{\eta_{\text{exp}}} \right| \times 100$) were evaluated for all test mixtures (i.e., those withheld from training and validation of the models), across the full range of solution complexities and the 55 °C range of temperatures investigated. In Figs. 3(a)–3(c), these residuals are binned by the number of components in solution, and in Figs. 3(d)–3(f), the percent errors are plotted as a function of the calculated size-dependent or Flory-style entropy of mixing $S_{\text{mix}} = -R \sum_{i=1}^n x_i \ln \phi_i$, wherein x_i gives the mole fraction of each component, ϕ_i gives the volume fraction of each component, and R is the ideal gas constant. We note that this form of the entropy of mixing has been found to better represent the mixing entropy of organic and aqueous solutions,^{46,47} but that use of the ideal entropy of mixing ($S_{\text{mix}} = -R \sum_{i=1}^n x_i \ln x_i$) here yields nearly identical insights.

Figure 3 reveals several key distinctions between the PINN, the KC, and the AAG models. First, the PINN model displays no significant trend in residuals vs solution complexity and no discernible bias toward under- or overprediction, including as a function of temperature. PINN residuals cluster closely around 0, with the average

residual less than 0.04 log(cP) in magnitude. This stability across conditions demonstrates the interpolability and extrapolability of the model. Conversely, both the ideal KC model and the AAG model show divergence of error as mixture complexity increases, in addition to a systematic bias toward significant underprediction. The ideal KC model, consistent with expectations given its complexity, yields both lower accuracy and reduced residual stability as solution complexity increases. We further note the clear relative superiority of the AAG model in capturing temperature dependence (as compared to the ideal KC), as evidenced by the tighter clustering of residuals across temperature for a given solution, and that the AAG and KC models experience opposite biases (the KC increasingly inaccurate at low temperature, the AAG increasingly accurate at the same). This suggests, consistent with prior observation,⁴⁸ that the AAG approach to scaling the viscosity at low temperature (which leverages the glass transition as a finite bound) is much more physically reasonable than any unbounded Arrhenius-style approach.

Critically, the trends in error observed with complexity in these canonical physical models hint at potential mechanisms contributing to the viscosity of many-component solutions that transcend

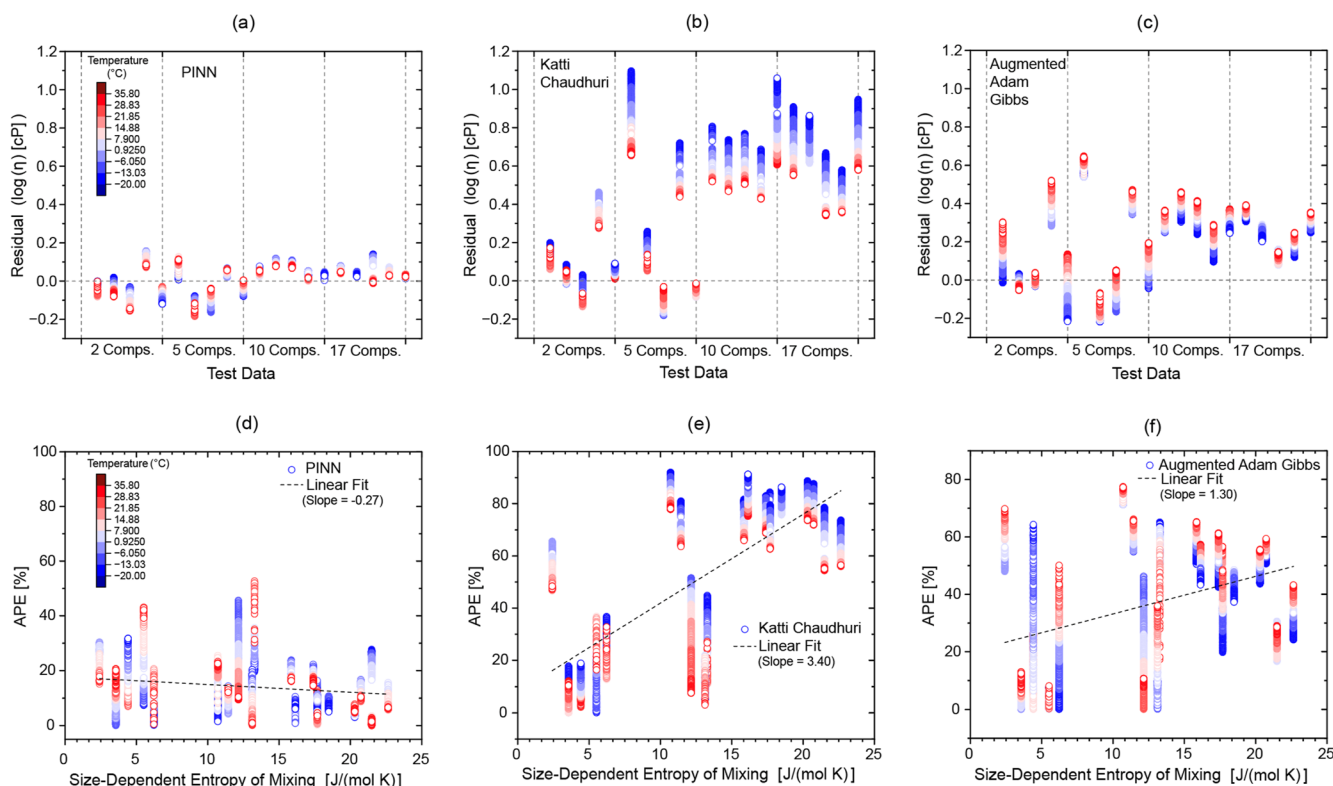


FIG. 3. Accuracy of different models as a function of solution complexity. (a)–(c) Residual distributions of predicted logarithmic viscosities across test mixtures of varying complexity (2–17 components) and temperatures (–20 to 35 °C), for the PINN, Katti–Chaudhuri, and AAG models. (d)–(f) Absolute percent error (APE) of viscosity predictions as a function of size-dependent (Flory-style) entropy of mixing for the PINN, KC, and AAG models. We note that while scatter in panels (d)–(f) may appear higher at the cold and warm extremes relative to panels (a)–(c), this is due to the nature of the absolute percent error measurement, which is a linear calculation that can produce different percent errors per unit viscosity at the temperature extremes where viscosity is changing logarithmically. Log-residuals in panels (a)–(c) should be considered stronger practical markers of prediction variability with temperature.

those represented within them. Intuitively, and supported in part by the trends observed in Figs. 3(d)–3(f), we suggest that these mechanisms may be related to the increasing dominance of mixing entropy in many-component solutions. Configurational entropy has long been considered a foundational determinant of solution viscosity,^{48,49} and both computational and scaling arguments have suggested that the thermodynamic scale of the intrinsic configurational entropy of a pure liquid may be equaled or surpassed by mixing entropy even in a binary solution.^{50,51} This implies that entropic phenomena (such as viscosity) in many-component solutions may be increasingly dominated by mixing entropies that are insufficiently captured in current physical models. Future investigations should consider sparse regression principles or other techniques suited for

high-throughput exploration of correlative physical inputs to further interrogate possible mechanisms at play.

Representative viscosity predictions and ViscoPINN application interface

To maximize the utility of the dataset and PINN model produced herein, we developed a user-friendly, MATLAB-based computer application (dubbed “ViscoPINN”) that uses the model to enable temperature-dependent viscosity prediction for arbitrary combinations of the components in Table I, between -20 and 35°C . In Figs. 4(a)–4(d), we show four representative test solutions, containing 2, 5, 10, and 17 components, respectively,

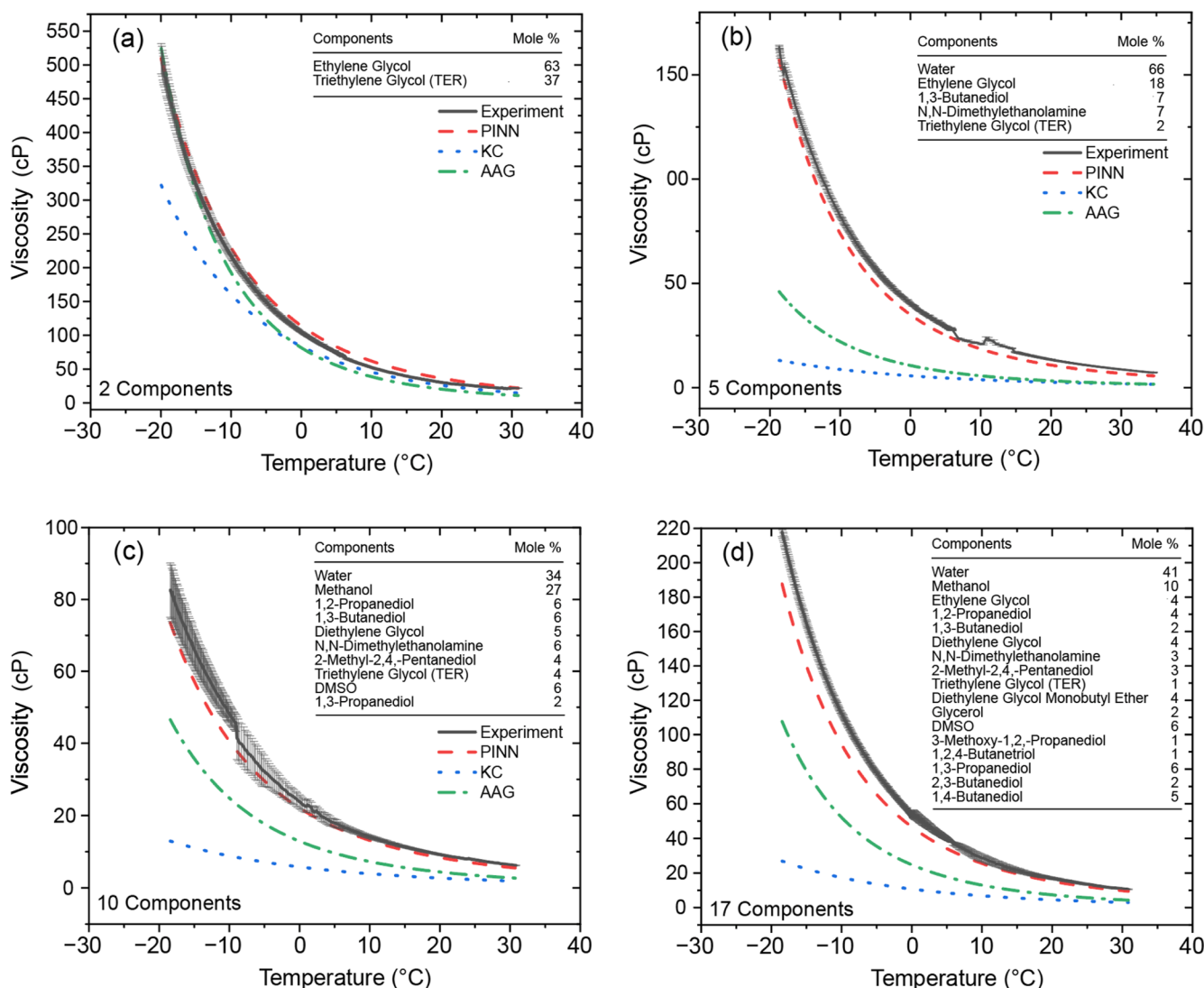


FIG. 4. Comparison of temperature-dependent viscosity predictions for representative test mixtures. Panels (a)–(d) show sample 2-, 5-, 10-, and 17-component solutions, respectively. Experimental data are compared with PINN, ideal Katti–Chaudhri, and augmented Adam–Gibbs models. Experimental uncertainty reflects the manufacturer-specified accuracy of the viscometer.

with the temperature-dependent viscosities predicted by the PINN, Katti–Chaudhri, and augmented Adam–Gibbs models compared against experimental data. Across these solutions, the model-to-model accuracy trends with temperature and solution complexity are well maintained, and the predictive power of the PINN model is illustrated.

An executable form of the ViscoPINN application is available in the [supplementary material](#), and a MATLAB-based implementation that offers direct function-level access to the trained PINN model, complemented by example scripts, is accessible through GitHub.⁵²

OUTLOOK

Viscosity is a required parameter for even the most basic design and analysis of liquid heat and mass transfer processes, and improved understanding of (and predictive power over) temperature-dependent organic solution viscosities is integral to process enhancement across myriad industrial domains. In heat-transfer fluids commonly used as coolants, antifreezes, and secondary refrigerants (e.g., multi-glycol aqueous solutions), viscosity affects convective heat transfer, pressure drop, and pump efficiency,¹⁸ with lower viscosities generally facilitating better performance. In lubricants, sealants, personal care products, and other multifunctional fluids, viscosity may dictate load-bearing capacity, film stability, and frictional behavior.¹ In the cryopreservation of tissues, organs, and organisms,⁵³ viscosity dictates the deliverability of aqueous organic cryoprotectant solutions into complex biological matter (both by diffusion⁵⁴ and perfusion⁹), kinetic aspects of ice nucleation and growth,⁵⁵ and the process of vitrification (i.e., glass formation).^{9,56} Liquids applied to these various applications are almost invariably comprised of many chemical components and utilized over a wide range of temperatures. This compositional (and thermal) complexity, compared against the narrow compositional space described in the literature to date,^{12,57–59} highlights the pressing need for new datasets and models that interrogate many-component solutions.

To meet this need, we provide here 44 316 temperature-dependent viscosity measurements across 100 aqueous and organic solutions, spanning 2–17 components and –20 to 35 °C. We tested predictive implementations of the ideal Katti–Chaudhri and augmented Adam–Gibbs models against these data and show that both systematically underpredict viscosity in increasingly complex solutions. By integrating these classical models as physical regularizers within a physics-informed neural network (PINN), we developed a synergistic approach that encodes the physical insight captured in each while allowing the neural network to implicitly model uncaptured phenomena, interactions, and non-idealities. Compared to strong conventional ANNs, the PINN reduced test-set errors by 6%–7%, and lowered RMSE by more than 70% relative to the augmented Adam–Gibbs model. Residual and entropy-of-mixing analyses showed that the PINN preserved accuracy in both simple and highly complex solutions, achieving near-zero bias with solution complexity even during extrapolation.

The success of this framework highlights two key advances: (i) explicitly incorporating physical laws into machine learning models stabilizes extrapolation beyond the training data, and (ii) combining novel empirical data with classical theories reveals the limitations

of idealized models while leveraging their strengths to guide learning. Together, these findings establish physics-informed machine learning as a powerful approach for predicting the transport properties of many-component liquids, with implications for the design of cryoprotectants, solvents, lubricants, heat-transfer fluids, and beyond.

SUPPLEMENTARY MATERIAL

[Supplementary Material](#) 1 includes additional methodological, data analysis, and uncertainty details. [Supplementary Material](#) 2, which is provided in both machine-readable spreadsheet form and Microsoft Word document form, includes the multi-component, temperature-dependent viscosity dataset, organized by solution composition. A supplementary compressed file, “ViscoPINN_Application_Windows_Executable.zip,” is also provided, which contains an executable Windows form of the ViscoPINN viscosity prediction application.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

S.K. and M.J.P.P. conceived the study, analyzed the results, and wrote the manuscript. S.K. performed the experiments with assistance from A.Z., and developed all models, software, and curated the dataset. A.Z. performed the uncertainty analysis. All authors reviewed, edited, and approved the final form of the manuscript.

Soheil Kavian: Conceptualization (supporting); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Arian Zarriz:** Data curation (supporting); Formal analysis (supporting); Methodology (supporting); Validation (supporting); Writing – review & editing (supporting). **Matthew J. Powell-Palm:** Conceptualization (lead); Formal analysis (supporting); Funding acquisition (lead); Investigation (supporting); Supervision (lead); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

All data presented in this work are provided in both document form and machine-readable spreadsheet form in the [supplementary](#)

material. The dataset and software described herein will also be posted to Mendeley Data with a permanent DOI, following peer review of this manuscript.

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